

CURRICULUM VITAE

Andrew, Chi-Chung Cheng

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 <https://andrewcheng2016.netlify.app/>

EDUCATION

- 2021-2023** **The Hong Kong Polytechnic University**
Research Postgraduate (Incomplete)
- 2021** **The Hong Kong Polytechnic University**
BEng (1st Hons) Electronic & Information Engineering

WORK EXPERIENCE

From	To	Position
Sep 2023	Now	Department of Computer Science, Hong Kong Chu Hai College Research Assistant
Sep 2021	Sep 2023	Department of Electronic and Information Engineering, The Hong Kong Polytechnic University Teaching Assistant
Jul 2020	Aug 2020	Electrical and Mechanical Services Department Summer Intern

PROFESSIONAL POSITIONS/SERVICES

From	To	Position
Apr 2024	Mar 2026	Faculty of Dentistry, The University of Hong Kong Honorary Research Associate in Restorative Dental Sciences

AWARDS AND SCHOLARSHIP

- 2020/2021** Dean's Honours List
- 2019/2020** The City Lions Club of Hong Kong Scholarship
- 2019/2020** Dean's Honours List
- 2019** Best GPA Award
- 2019** Best Semester GPA Award
- 2018** Best Semester GPA Award

RESEARCH INTERESTS

Artificial Intelligence, Deep Learning, Machine Learning, Interpretable Neural Networks, Image Segmentation

Technical Skills

Programming & Frameworks

- Proficient** React.js, Django, Flask API, Python, HTML, CSS, JavaScript, PyQt5
- Experienced** C++, C#, Unity, Java

Cloud & Machine Learning

- Proficient** Google Cloud Platform (Cloud Run, Vertex AI), TensorFlow, PyTorch

Languages

- Native** Cantonese
- Fluent** English, Mandarin

PROJECT EXPERIENCE

Hong Kong Chu Hai College

- Research on image segmentation for gum disease detection (2023-2025)

The Hong Kong Polytechnic University

- Research on interpretable neural networks via principal component analysis (2021-2023, as Research Postgraduate)
- Active noise cancellation with convolutional neural networks (2021, Undergraduate Final Year Project)

Electrical and Mechanical Services Department

- Developed IoT project using Arduino (ESP32) with mobile-controlled DHT11 sensors via Blynk cloud. (2020, as Summer Intern)

Publication and Patent List for Andrew, Chi-Chung Cheng

SUMMARY

My publications covering the following areas:

1. Image Segmentation.
2. Computer Vision.
3. Artificial Intelligence.

PUBLICATIONS & PATENTS

1. Reinhard Chun Wang Chau, Zhuohong Gong, Andrew Chi Chung Cheng, Kaijing Mao, Khaing Myat Thu, Ruonan Zhang, Tien Hsing Chang, Hong Jin Tan, Harris Sik-Ho Tsang, Maurizio S. Tonetti, Falk Schwendicke, Richard Tai-Chiu Hsung, Walter Yu Hang Lam, "External Validation of a mHealth Tool for Detecting Gingival Inflammation in Community-Dwelling Older Adults," submitted to Journal of Dentistry (Feb 13, 2026)
2. Chi-Chung Cheng, Harris Sik-Ho Tsang, Richard Tai-Chiu Hsung, Yui-Lam Chan, Wai-Lun Lo and Walter Lam, "GumAgent: Towards an Accessible Gum Disease Detection Tool Leveraging Vision Language Model," *The 8th International Conference on Information Communication and Signal Processing (ICICSP 2025)*
3. Reinhard Chun Wang Chau, Andrew Chi Chung Cheng, Kaijing Mao, Khaing Myat Thu, Zhaoting Ling, In Meei Tew, Tien Hsin Chang, Hong Jin Tan, Colman McGrath, Wai-Lun Lo, et al., "External validation of an AI mHealth tool for gingivitis detection among older adults at daycare centers: a pilot study," *International Dental Journal*, 2025.
4. Andrew Chi-Chung Cheng, Richard Tai-Chiu Hsung, Guan-Hua Li, In Meei Tew, Wai-Lun Lo, and Walter Yu Hang Lam, "Developing automated photographic detection of gum diseases using deep neural networks for mobile devices," 2024 6th International Conference on Software Engineering and Development (ICSED2024), May 29-31, 2024, Hong Kong, <https://doi.org/10.1145/3686614.3686621>.
5. Yu Hang LAM; Richard Tai Chiu HSUNG; Andrew Chi Chung CHENG; Guan-Hua LI; Wai-Lun LO; Chun Wang CHAU; and Khaing Myat THU, "Machine Learning-Based Image Processor For Intraoral Images", (2024) US patent application number 18/608,983. Filing Date: 03/19/2024.
6. Yu Hang LAM; Tai Chiu HSUNG; Chi Chung CHENG; Sik Ho Tsang; Wai Lun LO and Kai Jing MAO, "DTCWT based Shift-Equivariant pooling for medical image segmentation", (2026) US patent application number 63/957,242. Filing Date: 09/01/2026.

OTHER GRANTS:

1. Reinhard Chun Wang Chau, Walter Yu Hang Lam, Richard Tai Chiu Hsung, Nelson Chun Yin Chau, Khaing Myat Thu, Andrew Chi Chung Cheng, Kaijing Mao; *Google Cloud x HKU Startup Connector*, Title: "Gingivitis Detection Solution", Duration: Jun 2024 - May 2026. Total Amount granted: USD250,000.